

GABRIEL POPA

Senior Software Engineer - Vue • Node (Express/Nest) • AWS

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ABOUT ME

Senior Software Engineer with **11+ years of experience** building scalable **Vue.js** frontends and reliable **Node.js** backend APIs for modern web platforms, turning complex and evolving requirements into stable production releases. Strong expertise in **JavaScript, TypeScript, Vue 2/3, Node.js, REST/GraphQL APIs, PostgreSQL, Redis, Kafka, WebSockets**, and **AWS**, with hands-on experience designing maintainable services and resilient client applications. Known for modernizing legacy systems by migrating older frontend patterns into more maintainable **component-based Vue architectures**, safely upgrading **Node.js** runtimes, and resolving high-impact issues such as race conditions, memory leaks, and unstable builds through profiling, testing, performance tuning, and controlled rollout strategies.

SKILLS

- **Frontend: Vue.js** 2–3, JavaScript, TypeScript, **Nuxt.js** basics, Pinia, Vuex, Vue Router, Composition API, Options API, reusable component architecture, Vite, Webpack, Jest, Vue Test Utils, Cypress, Playwright, HTML5, CSS3, Tailwind CSS, responsive UI, SPA architecture, component design patterns
- **Backend & APIs: Node.js** 12–20, Express.js, NestJS, REST APIs, GraphQL, gRPC basics, WebSockets, event-driven services, JWT/OAuth2 authentication, OpenAPI/Swagger, background workers, job queues, API versioning, middleware architecture
- **Data / Messaging / Search:** PostgreSQL, MySQL, Redis, Kafka, RabbitMQ, Elasticsearch/OpenSearch basics, query optimization, indexing strategies, schema design, event streaming, pub/sub messaging
- **Cloud & DevOps:** AWS (EC2, S3, RDS, ECS/EKS basics, IAM basics), Docker, Terraform basics, GitHub Actions, GitLab CI, CI/CD pipelines, Nginx, Linux, containerized deployments, environment configuration, build pipelines
- **Observability / Quality / Security:** OpenTelemetry basics, Prometheus/Grafana basics, Sentry, structured logging, distributed tracing basics, rate limiting, OWASP security practices, Jest, Vue Test Utils, Supertest, Cypress, Playwright, unit testing, integration testing, contract testing, end-to-end testing, error monitoring

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Soft Build | Bucharest, Romania (Jan2024 – Dec 2025)

- Built and led development of a multi-tenant dashboard using **Vue 3, TypeScript**, and modular frontend patterns, refactoring unstable screens into reusable components that improved UI consistency and reduced tenant-specific issues.
- Improved frontend performance by profiling slow-rendering views, reducing unnecessary reactive updates, and introducing smarter client-side caching, which reduced interface freezes and kept large tables and filters responsive.
- Modernized frontend architecture by adopting the **Composition API** for complex feature areas, reducing component complexity and making shared business logic easier to test and reuse.
- Enhanced real-time user experiences by improving **WebSocket** reliability with reconnection handling and buffered event queues, keeping live status updates stable during temporary network disruptions.
- Built scalable backend services with **Node.js, NestJS**, and **TypeScript**, delivering maintainable **REST** and **GraphQL** APIs with clear service boundaries and strong modular design.
- Modernized a legacy **Node.js** API from v14 to v20 through staged upgrades, compatibility validation, and safe deployment planning, avoiding disruption to critical business workflows.
- Resolved a **production memory leak** by identifying faults in in-memory caching and stale event listeners through heap analysis and load testing, reducing memory-related incidents by **46%** while restoring service stability under sustained traffic.

- Prevented duplicate transaction processing by implementing idempotency controls, strengthening **PostgreSQL** constraints, and refining retry handling for downstream service failures.
- Strengthened application security by implementing **JWT** and **OAuth2** authentication, **RBAC** authorization, audit logging, and request rate limiting to improve access control and platform protection.
- Integrated **Kafka** and **RabbitMQ** for asynchronous event flows supporting notifications, export generation, and downstream status updates, reducing tight coupling by **38%** between user-triggered actions and background processing.
- Introduced **OpenAPI** contract testing into **GitHub Actions** CI/CD pipelines, enabling frontend and backend teams to release independently with lower integration risk.
- Improved production observability through structured logging, **Sentry**, and **OpenTelemetry** metrics sent to **Prometheus/Grafana**, making incidents easier to detect, trace, and resolve.
- Containerized backend services with **Docker** and deployed them on **AWS EC2/ECS**, automating build and release workflows through CI pipelines to improve deployment consistency.
- Implemented **Elasticsearch/OpenSearch**-based search capabilities, enabling fast tenant-scoped queries and scalable filtering across large operational datasets.

Software Engineer

Next Apps | Ghent, Belgium (*Nov2020 – Nov 2023*)

- Built customer-facing workflows with **Vue.js 2-3** and **TypeScript**, improving form reliability, validation consistency, and accessibility across complex product journeys.
- Partnered with design teams to create a reusable **Vue** component library, reducing duplicated UI work while supporting product-specific customization needs.
- Modernized legacy frontend modules by migrating heavier Options API logic into more maintainable shared composables where appropriate, improving code organization and reducing duplication.
- Improved frontend build tooling by simplifying heavy **Webpack** configuration and later adopting faster tooling patterns, which sped up local development and reduced CI build instability.
- Developed interactive dashboards with **Vue Router**, centralized state handling, and reusable form patterns, making navigation flows easier to maintain across large feature areas.
- Modernized a legacy **Express.js** API by migrating it to **NestJS**, reorganizing routes into modular controllers and services to improve maintainability and code structure.
- Developed **Node.js** services with versioned **REST** endpoints and deprecation strategies, supporting backward compatibility while reducing coupling between API consumers and backend changes.
- Implemented **Redis** caching for high-traffic endpoints using safe invalidation rules and TTL strategies, reducing database load while keeping data accurate.
- Resolved intermittent **502/504** errors by tuning **Nginx** timeout settings, optimizing slow database queries, and adding latency monitoring around critical user workflows.
- Fixed a **data consistency issue** caused by out-of-order event processing by enforcing ordering keys and adding a background reconciliation process, reducing record mismatches by **39%**.
- Improved asynchronous job reliability by adding retry policies, dead-letter handling, and monitoring dashboards, making failed or stuck workloads easier to detect and recover.
- Strengthened authentication flows by tightening token validation, adding rate limiting for abuse protection, and safely rotating secrets across services.
- Introduced structured logging and correlation IDs across frontend and backend systems, making it faster to trace production issues from user actions to backend handlers.
- Improved release reliability by enforcing code review checklists, maintaining test coverage thresholds, and documenting rollback steps for higher-risk production changes.

Software Developer

Datamart | Stockholm, Sweden (*Feb 2016 – Sept 2020*)

- Built internal analytics dashboards using **Vue.js** and **Node.js**, integrating large datasets into responsive interfaces with server-side pagination and caching to improve usability and performance.
- Supported the refactoring of a large **Node.js** API into smaller modular services behind an API gateway, improving maintainability while preserving stability for existing integrations.
- Improved **reporting performance** by investigating slow pages, adding targeted **PostgreSQL** indexes, rewriting heavy joins, and reviewing execution plans, which reduced query latency by **41%**.

- Modernized legacy frontend code by restructuring older view logic into reusable **Vue** components with clearer state boundaries, improving maintainability and reducing UI regressions.
- Resolved a **difficult data issue** by identifying timezone parsing differences between the **UI**, **API**, and database, then standardizing date handling, which restored reporting accuracy and reduced date-related reporting errors by **36%**.
- Implemented background job processing for large file exports, replacing synchronous requests that frequently timed out and adding status polling to improve user visibility.
- Added automated integration tests around core business rules, reducing regressions and helping maintain consistent behavior during frequent releases.
- Improved CI pipeline reliability by separating fast unit tests from slower integration tests, reducing flaky build failures and speeding up feedback cycles.
- Introduced audit logging for data update workflows, enabling teams to trace user actions and review change history during issue investigations.
- Reduced **frontend bundle size** by implementing **code splitting**, lazy-loaded route handling, and removing unused dependencies, improving page load times and overall client-side performance by **34%**.
- Assisted with production releases by introducing smoke tests and staged deployments, helping identify configuration issues before they affected users.

Junior Software Developer

Berg Software | Timișoara, Romania (Sep2014 – Mar 2016)

- Built early SPA modules using **Node.js** and **Express**, creating **REST** endpoints with consistent response handling and basic error management across backend workflows.
- **Improved legacy web pages** by introducing modular **JavaScript**, stronger client-side validation, and clearer form feedback, reducing user input errors and related support requests by **28%**.
- Helped resolve a recurring logout issue by fixing session cookie settings and aligning server-side session timeouts with browser behavior, improving login reliability for users.
- Developed reusable UI components and shared helper utilities, reducing duplicated code and making common frontend updates faster and easier for the team.
- Improved **page load performance** by enabling asset compression, configuring browser caching, and removing blocking scripts that delayed initial rendering, cutting initial load time by **32%**.
- Supported deployment automation by writing simple release scripts and maintaining deployment documentation, helping the team deliver updates more safely and consistently.
- Assisted with production issue resolution by reviewing logs, reproducing bugs locally, and applying targeted fixes under guidance from senior engineers.
- Improved **database performance** by writing safer **SQL** queries and adding basic indexes to support admin dashboards as data volume increased, reducing query response time by **29%**.
- Used a structured debugging approach by documenting reproduction steps, reviewing logs, and delivering small, low-risk fixes instead of large refactors.

EDUCATION

Bachelor's Degree in Computer Science

University of Bucharest | (2010 – 2014)

- Focused on practical software engineering fundamentals that translated directly into building reliable web systems and data-driven applications.

CERTIFICATIONS

AWS Certified Developer – Associate — 2024

- Demonstrated practical capability in building, deploying, and maintaining **Node.js-based backend services** on **AWS**, with focus on service integration, monitoring, secure configuration, and production-ready cloud operations.

OpenJS Node.js Application Developer (JSNAD) — 2023

- Validated hands-on understanding of **Node.js** application development, including asynchronous programming, package management, debugging, testing, and building maintainable backend services with modern JavaScript patterns.

PostgreSQL for Developers — 2022

- Demonstrated working knowledge of relational modeling, query optimization, indexing strategies, transaction-aware data access, and performance-focused backend persistence patterns commonly used in **API-driven Node.js platforms**.